Fiscal Year:	FY 2018	**Task Last Updated:**	FY 02/16/2018
PI Name:	Long, Marshall Ph.D.		
Project Title:	Fuel Dilution Studies of Coflow Laminar Diffusion Flames in a Microgravity Environment		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	06520-8284	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	03/01/2017	**End Date:**	02/28/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	NOTE: Period of performance changed to 3/1/2017-2/28/2020 (original POP was 1/1/2017-12/31/2019) per D. Stocker/GRC (Ed., 1/17/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Smooke, Mitchell Ph.D. (Yale University)		
Grant/Contract No.:	NNX17AD49A		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: Continuation of "Coflow Laminar Diffusion Flame" grant NNX11AP43A, with investigators Drs. Marshall Long and Mitchell Smooke. Diffusion flames are the flame type of most practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in such systems is critical in the modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation, and radiative transfer take place. Turbulent diffusion flames are generally nonbuoyant due to the large flow velocities. In addition, in practical devices combustor configuration and flow field patterns may produce flames with sharp edges. Extinction along these edges can result from incomplete combustion. In other regions combustion generated soot particulates may be formed. To understand better the factors that affect flame extinction and particulate formation, we propose to focus this study on		

Task Description:	flame phenomena at the extremes of the fuel dilution spectrum – namely dilute flames near extinction and pure-fuel (sooting) flames. Previous work has shown that existing computational models are unable to predict accurately the behavior of these systems. The enhanced time and spatial scales in a microgravity environment will enable the stabilization of flames with increased levels of dilution and will help generate flames that have increased soot residence times compared to normal gravity facilities. This should provide an excellent platform to gain a better understanding of the factors that affect diffusion flame lift-off distance and extinction, and it should also provide an excellent test-bed to determine the roles of inception, surface growth, and oxidation processes in soot formation models. The overall goals of the research are to improve our understanding of the physical and chemical processes controlling diffusion (i.e., non-premixed) flame structure and lifting phenomena (i.e., stabilization) and to provide for rigorous testing of numerical models, including thermal radiation, soot formation, and detailed chemical kinetics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Combustion of fossil fuels is responsible for the majority of energy production in the world today and provides the power for transportation systems of all kinds. Effective modeling of combustion processes is critical to improving efficiencies and reducing pollutants in practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in diffusion flames will improve modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation, and radiative transfer take place.
Task Progress:	Microgravity Experiments: ACME and CLD Flame The Coflow Laminar Diffusion (CLD) Flame experiment is one of the experiments in the Advanced Combustion via Microgravity Experiments (ACME), which began in late 2017. Much of our effort during the first part of the award period concentrated on preparation for the start of ACME and CLD Flame. Since ours is the first of five experiments that will use the new ACME insert, considerable time was spent on refinement and augmentation of the CLD Flame test matrices to include initial testing of the ACME insert. The preliminary ACME scripts were reviewed and, when possible, preliminary hardware settings were suggested based on normal gravity results. In June, Davide Giassi and Marshall Long spent two days at NASA Glenn taking part in tests of the ground copy of the ACME insert. Sample data from the tests were made available on a NASA server and analysis of that data began. Processing software to handle data provided by ACME was refined. Tests mimicking those planned for the flight experiments showed that one of the mass flow controllers was prone to oscillations. This turned out to be an issue with the flight hardware as well. In the time leading up to initial testing, refinements to the test matrix for CLD Flame were made, with spreadsheets updated to include graphical representations of the timing of flow changes and to contain entries needed for the scripts that will control the experiment. By September, updated test matrices for initial ACME insert testing and CLD Flame experiments were provided to NASA. Different scenarios were established for initial camera testing, ignition testing, fuel dilution testing, flow ramping, and CLD Flame operational tests. A Ground-Based Copy of CLD Flame In order to provide data under normal gravity for comparison with the microgravity results, as well as to help in planning tests on the International Space Station (ISS), hardware has been developed that closely mimics that of ACME. The ground copy of the CLD Flame burner, which was previously characterized in our laboratory, is enclosed in a pressure vessel. A color CCD camera and zoom lens system of the same make and model as that utilized in the ACME insert are used to collect images. Specific steps taken during this funding period to allow ground-based CLD Flame measurements in hardware as close to that of ACME as possible include the following: • The chamber volume was increased to ~82 liters to mimic that of the Combustion Integrated Rack (CIR), which houses the ACME insert. • Additional software was developed to remotely control the mass flow controllers (MFCs) and simulate the flow ramping that will characterize the ACME experiments. The software runs as part of the OMA program, which we use for data acquisition and analysis and has been made available to all of the ACME experiments. A custom circuit was built to interface OMA and the MFCs. • An igniter assembly, with a remotely operated arm, was manufactured and placed inside the chamber. • Igniters similar to what will be used in ACME were fabricated and tested. Since no tests of the ability of the ACME igniters to light gaseous flames were done prior to testing on the ISS, a series of tests was undertaken to assess the ability of the igniters to light our flames, the voltage required to light different fuels (related to temperature and power), and their expected lifetimes. This latter is particularly important as changing igniters requires significant operational time, since the chamber must be opened. For our igniters, the minimum voltage required to ignite flames with different fuels was determined. By imaging the igniter and measuring the color ratios (which depend on the temperature) with the color camera, we have a way of comparing the behavior of the ACME igniters with ours and extrapolating the optimal ignition voltages for the various fuel dilutions. Furthermore, our tests indicate that the lifetime is sensitive to the time the igniter is exposed to flame. Normal Gravity Tests Using the Yale CLD Flame Rig Using the ground copy of the CLD Flame burner inside our pressure chamber, we have successfully performed normal gravity tests for the different scenarios included in the CLD Flame test matrices. In addition to acquiring images with our copy of the color data camera, telemetry similar to that returned by ACME are recorded. A video of a 40% methane flame test is available at the following url: http://guilford.eng.yale.edu/ Temperature measurements in selected non-sooty flames were done using a SiC fiber and employing thin filament pyrometry. These tests were useful in determining the effectiveness and accuracy of temperature from images taken without the blue-green (BG) color filter. Additionally, these tests confirmed the need for image defocus when thin

	filament pyrometry is implemented with the ACME setup. Specifically, images of nitrogen-diluted methane flames (stabilized on the CLD Flame burner, flow velocity in the range 10 to 35 cm/s, and fuel concentration ranging from 40% to 100%) were collected as a function of time and chamber pressure. Due to the relation between test duration and chamber pressure, this helped determine suitable temporally-stable flames whose temperature can be measured using thin filament pyrometry. While Phase One of the CLD Flame experiments will concentrate on weak, non-sooting flames, measurements were also made in sooting flames, which will be the focus of the second phase of the CLD Flame experiments. Images of nitrogen-diluted ethylene flames (stabilized on the CLD Flame burner, flow velocity in the range 10 to 50 cm/s, and fuel concentration ranging from 20% to 100%) were collected with the ACME color data camera without the use of the BG color filter to investigate the capability of the detector to return accurate soot temperature information when no filter is used. For selected flames, soot temperature was also evaluated with a Nikon D300s as a way to confirm the results. Computational and Experimental Study of the Effects of Pressure and Fuel Dilution Since the CLD Flame experiments are carried out in the enclosed pressure vessel, there is an increase in pressure during each test due to the addition of fresh gas into the chamber (primarily from the coflow). To understand better the effects of this increasing pressure, we have conducted a combined computational and experimental investigation to characterize the influences of pressure, as well as fuel stream dilution on the structure, shape, and sooting behavior of coflow CH4-air laminar diffusion flames. A series of 72 CH4-air coflow laminar diffusion flames with four levels of N2 dilution in the fuel stream (i.e., 50, 55, 60, and 65% CH4) and 18 levels of pressure (i.e., 1.0 to 2.7 atm in 0.1 atm increments) were assessed. In general, very good agreement between the computations and the experiments was achieved for the temperature, the flame shape, and the soot volume fraction, and the numerical model successfully captured the effects of varying pressure and fuel stream dilution. The major conclusions of this research were as follows: • The maximum temperature increases with increasing CH4 concentration or pressure. • The lift-off height decreases significantly with increasing pressure, the modified flame length is roughly independent of pressure, and the flame radius decreases with pressure. • The pressure and fuel stream dilution significantly affect the spatial distribution and the peak value of the soot volume fraction. Results of this study were published in Combustion Theory and Modelling.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Cao S, Ma B, Giassi D, Bennett BAV, Long MB, Smooke MD. "Effects of pressure and fuel dilution on coflow laminar methane-air diffusion flames: A computational and experimental study." Combustion Theory and Modelling. 2018;22(2):316-37. Published online: 13 Dec 2017. https://doi.org/10.1080/13647830.2017.1403051 , Jan-2018
Dissertations and Theses	Giassi D. "Optical Diagnostics Applied to Quantitative Characterization of Coflow Laminar Diffusion Flames in Microgravity and Normal Gravity." Ph.D. Thesis, Yale University, May 2017. http://guilford.eng.yale.edu/davideGiassiPhdThesis.pdf , May-2017